

SYSTEMATIC REVIEW

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The effect of virtual reality stimulation on reducing the incidence of delirium among patients admitted to intensive care units: a systematic review study

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Abstract

Introduction The intensive care unit (ICU) provides critical care for severely ill patients, often leading to prolonged stays and psychological issues like delirium, affecting 35%-80% of patients. Delirium impacts cognition and increases mortality. Virtual Reality (VR) shows potential for alleviating symptoms but lacks conclusive evidence for reducing delirium, necessitating further research. The main goal of this systematic review is to evaluate whether VR stimulation can reduce the incidence of delirium among critically ill patients.

Methods This systematic review follows PRISMA guidelines. Studies on VR for delirium prevention in adult ICU patients up to January 2024 were searched in online databases. Inclusion criteria focused on English-language studies specifically addressing VR's role in delirium prevention. The review excluded non-relevant studies and assessed quality using the ROBINS-I tool for bias in non-randomized studies. Data were synthesized qualitatively, summarizing key findings and evaluating patient perceptions of VR use in ICU settings. The study is registered on PROSPERO (CRD42024505083).

Results The findings indicate that VR may help reduce delirium incidence by providing cognitive stimulation and alleviating sensory overload, with some studies reporting zero rates of delirium. Patients experienced relaxing effects on vital signs, improved pain management, and found VR sessions enjoyable and tolerable, while providers noted ease of use. Importantly, no significant adverse events were reported, and VR interventions can be adapted for infection control in ICU settings. Nature based VR environments were particularly effective in promoting relaxation compared to urban settings, highlighting VR's potential as a safe and beneficial tool for critically ill patients, although these observations are based on a small number of heterogeneous studies with varying designs, populations, and intervention protocols.

Conclusion The findings support VR as a practical and safe digital therapy in ICUs, though further evaluation of its role in delirium prevention is needed.

Keywords Delirium, Critical care, Virtual reality

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Contribution of the paper

What is already known about the topic:

- Delirium affects 35%–80% of ICU patients, leading to cognitive impairment and increased mortality.
- Traditional pharmacological treatments for delirium often lack efficacy and can have adverse effects.
- Non-pharmacological strategies, while recommended, have limited clinical evidence supporting their effectiveness.

What this paper adds:

- This systematic review provides evidence that Virtual Reality (VR) may reduce the incidence of delirium in ICU patients by offering cognitive stimulation and alleviating sensory overload.
- The findings suggest that VR interventions seem safe, well-tolerated, and enjoyable for critically ill patients, with no significant adverse events reported.
- Nature-based VR environments were particularly effective in promoting relaxation and reducing physiological stress responses, indicating a novel approach to delirium prevention in ICU settings.

Introduction

Delirium is one of the most frequent and serious complications affecting patients in intensive care units (ICUs), with prevalence rates ranging from 35% to 80% among both ventilated and non-ventilated critically ill patients [1]. It is a complex neuropsychiatric syndrome characterized by disturbances in attention, cognition, and awareness, often fluctuating over time [2]. Delirium in ICU patients is associated with numerous adverse outcomes, including increased mortality, prolonged mechanical ventilation, unplanned device removal, and longer ICU and hospital stays [3]. Importantly, delirium also contributes to long-term cognitive impairments that affect up to 75% of ICU survivors, negatively impacting memory, attention, information processing, and executive function [4–9].

The ICU environment itself can exacerbate delirium risk by presenting patients with both sensory overload and sensory deprivation, leading to cognitive and attentional fatigue [10–15]. Critically ill patients are exposed to frequent alarms, bright lights, noise, and interruptions, yet simultaneously face social isolation and disorientation due to sedation and limited mobility [12]. These factors combine to increase vulnerability to delirium, which may manifest in hyperactive, hypoactive, or mixed forms [6].

Currently, prevention and management strategies include pharmacological and non-pharmacological approaches. Pharmacological treatments such as

antipsychotics and dexmedetomidine are used to control agitation and distress but have not demonstrated consistent effectiveness in reducing delirium duration or improving clinical outcomes [16]. Clinical guidelines generally recommend against routine use of these drugs except in specific cases of distress or agitation unrelated to delirium [17].

Non-pharmacological interventions, although supported mainly by preliminary and inconsistent clinical trial evidence, are widely employed. These include early mobilization, sleep hygiene promotion, minimizing sedatives, psychological support, and patient reorientation [17–19]. Recent guidelines advocate the development and application of non-pharmacological therapies to prevent and manage delirium [17, 20].

Among emerging interventions, Virtual Reality (VR) offers a novel approach. VR provides immersive, computer-generated 3D environments that can stimulate multiple senses synchronously, often using head-mounted displays to fully engage the user's visual field [21, 22].

VR systems exist along a spectrum of immersion. Fully immersive VR employs head mounted displays (HMDs) with stereoscopic visuals, wide field of view optics, and head tracking sensors to block real world stimuli and create a strong sense of presence in the simulated environment [23, 24]. Semi-immersive configurations, including large projection screens or multi panel displays, partially replace the user's natural sensory input while maintaining some environmental awareness, which can be advantageous in clinical monitoring contexts [25]. Non-immersive formats, often delivered via standard monitors, provide virtual interaction without the enveloping sensory substitution of immersive systems. Augmented reality (AR) and mixed reality (MR) blend real and virtual elements, allowing users to interact with both simultaneously, a property potentially useful in ICUs where patient awareness of clinical surroundings is beneficial [26, 27]. Across all modalities, the central principle is delivering controlled, interactive, computer generated sensory stimuli to engage cognitive and perceptual processes relevant to therapeutic goals.

In healthcare, VR is increasingly applied for pain management, anxiety reduction, cognitive rehabilitation, and relaxation therapy [28–30]. Specifically in the ICU, VR has been explored as a meditation and distraction tool to help patients cope with the stressful environment by delivering calming content such as nature scenes, music, or guided exercises [29, 30]. VR requires minimal cognitive effort while promoting positive emotional effects such as immediate stress reduction and increased feelings of well-being [31].

Compared with other non-pharmacological interventions, VR presents several unique advantages that

justify its selection for investigation in ICU settings. Traditional interventions such as early mobilization, family engagement, or structured reorientation are often constrained by staffing limitations, infection control policies, patient immobility, and the frequent need for sedation [17, 32]. Likewise, approaches such as music therapy or environmental modifications may provide partial benefits but are less immersive and rely on sustained attention, which is often impaired in critically ill patients [33].

Similarly, sleep hygiene protocols and reorientation strategies may be disrupted by the ICU's noisy and unpredictable environment, with inconsistent effects on delirium incidence [35]. Psychological support, while beneficial, often requires active patient participation, which may not be possible for sedated or cognitively impaired individuals [34–36].

In contrast, VR delivers a fully immersive experience that requires minimal cognitive or physical effort from the patient, making it suitable even for those who are bedbound or sedated at light to moderate levels [36]. Moreover, VR can simultaneously target multiple delirium related risk factors such as anxiety, pain, sleep disruption, and sensory dysregulation through controlled, multisensory input [31, 37, 38]. By providing immersive environments that mitigate sensory overload and promote relaxation, VR addresses multiple delirium risk factors simultaneously, including stress, disorientation, and poor sleep quality, making it a promising candidate for further exploration in ICU settings [39]. Feasibility studies demonstrate that VR is generally safe, acceptable to patients, and adaptable within ICU environments [36, 39]. These characteristics make VR a particularly promising and practical candidate among non-pharmacological interventions for delirium prevention.

Importantly, previous studies have shown that VR may help reduce key delirium risk factors in ICU patients, including pain, poor sleep quality, and cognitive decline [38, 40, 41]. By reducing sensory overload and providing controlled sensory input, VR may help maintain cognitive function and potentially reduce delirium incidence [42, 43]. However, preliminary evidence suggests potential benefits of VR in preventing delirium among critically ill patients, though the evidence remains inconsistent and requires further investigation [36, 44].

Given the burden of delirium and the urgent need for effective preventive strategies, it is essential to systematically evaluate the role of VR interventions in this context. The purpose of this review is to explore the potential of VR stimulation to reduce delirium incidence in critically ill patients in ICU.

Method

Literature search

This systematic review was reported based on the PRISMA guideline [45]. A systematic literature search was conducted across three major databases: PubMed, Scopus, and Web of Science. Although other databases such as Cochrane Library, Embase, or CINAHL may also provide relevant studies, substantial overlap in indexing exists across these sources. The selected databases were also selected for their broad coverage of biomedical, clinical, and technological research relevant to virtual reality and ICU delirium interventions. To further reduce the risk of missing eligible studies, reference list screening of included articles and relevant reviews was also conducted.

No restrictions were placed on publication date to capture all relevant studies up to January 2024. To maximize search sensitivity and minimize bias, a combination of MeSH terms and free-text keywords was used. The search strategy incorporated synonyms and related terms to capture all relevant studies. Although prevention related keywords such as “prevent,” “incidence,” and “reduce” were not explicitly included, the combination of “delirium,” “ICU/critical care,” and “virtual reality” terms was used to capture all relevant studies on both prevention and management. Studies that investigated delirium incidence or reduction were identified within this search framework.

The following search strategy was used: (((Virtual Reality) OR (VR therapy) OR (Simulated reality) OR (VR stimulation)) AND (Delirium)) AND ((Intensive Care Unit) OR (critical care unit)).

Additional studies were identified through screening of all included studies and relevant reviews' reference lists. Duplicate removal was undertaken using EndNote.

Eligibility and selection criteria

Inclusion criteria included: (1) Studies published in English, (2) Studies involving adult patients (≥ 18 years) admitted to ICU or CCU, and (3) Studies investigating the use VR interventions specifically aimed at delirium prevention.

For the purposes of this review, VR interventions were defined broadly to include immersive systems (e.g., head-mounted displays), semi-immersive systems (e.g., projection based or large screen simulations), and augmented or mixed-reality hybrids, provided the intervention created an interactive computer generated environment intended to engage sensory and cognitive processes relevant to delirium prevention; this inclusive definition was chosen to capture the full technological spectrum used in ICU settings.

Exclusion criteria included: Studies involving pediatric patients (< 18 years), Studies conducted in non-ICU/

CCU settings, Studies using VR for outcomes unrelated to delirium prevention (e.g., pain management without delirium outcomes), Studies employing interventions for delirium prevention other than VR, Conference abstracts, study protocols, reviews, editorials, and opinion papers.

Review articles were screened for potentially eligible primary studies.

Study selection occurred through two stages. First, two independent reviewers (MS+ZSM) screened Title and abstracts of articles independently for eligibility using EndNote software and then relevant studies were examined in full-text and assessed in detail for compliance with eligibility criteria by two reviewers (MS+ZSM). An independent Third investigator (AH) was involved in the case of discrepancies in the extraction and assessment of the data. The following data were abstracted: author's name, year of publication, country, population, aim of the study, study type, Intervention type, Comparator (if applicable), and outcomes (including Acceptability).

The study was registered on PROSPERO (CRD42024505083).

Quality assessment

Two authors (MS and ZSM) independently evaluated the risk of bias for each included study, with disagreements resolved by consulting a third author (AH). Given the absence of randomized controlled trials (RCTs) in this emerging field, the Risk of Bias in Non-Randomized Studies of Interventions (ROBINS-I) tool was used to assess study quality. While the GRADE framework is widely recommended for rating the certainty of evidence, it is primarily structured for randomized controlled trials; therefore, its direct application was not feasible given our exclusively non-randomized evidence base. Instead, principles of transparent certainty assessment were incorporated by using ROBINS-I result to guide the interpretation of findings. This tool evaluates risk of bias across seven domains: confounding, participant selection, classification of interventions, deviations from intended interventions, missing data, outcome measurement, and selective reporting. Risk of bias was categorized as low, moderate, serious, or critical [46]. Studies

were not excluded based solely on risk of bias but the quality assessment informed the interpretation of results and strength of evidence.

Data synthesis

Key characteristics and outcomes of included studies were summarized (Table 1) and qualitatively synthesized based on the clinical focus of each study. Patient acceptability and perceptions regarding VR use during ICU/CCU stays were also reviewed narratively to provide insight into feasibility and user experience.

Results

The initial search identified 395 articles. After removing duplicates and screening titles and abstracts based on predefined inclusion/exclusion criteria, 68 studies underwent full-text review. 59 studies were excluded primarily due to targeting VR effects on outcomes other than delirium prevention, study design inconsistencies, non-ICU/CCU populations, or use of interventions other than VR. Finally 8 studies were included in the study (see Fig. 1 for PRISMA flow diagram [45]).

All included studies were published from 2017 to 2023 [42, 47–50] (Table 2). The country of study varied with studies taking place in Switzerland [36, 48, 51], China [52], USA [47, 49, 50], and Spain [42]. Participant ages ranged from 20 to 87 years. Methodological quality assessments using the ROBINS-I tool are summarized in Table 1; and Fig. 2.

Results of the included studies suggest that there are four aspects that can be considered when assessing the effectiveness of VR for delirium prevention among ICU patients. Based on that, this section is divided into four categories reviewing Efficacy of VR Interventions, patient Outcomes, adverse Events and subgroup analyses.

Efficacy of VR interventions

This section summarizes the findings related to the effectiveness of VR interventions in preventing delirium in ICU patients.

Providing visual and auditory stimuli to critically ill patients are important for achievement of full recovery [48]. In Turon's pilot study, assessment of efficacy of daily

Table 1 ROBINS-I risk of bias summary

Study	Confounding	Selection	Classification	Deviations	Missing data	Measurement	Selection reported	Overall
He 2021 [52]	Moderate	Low	Low	Low	Low	Low	Low	Moderate
Jawed 2021 [47]	Low	Low	Low	Moderate	Low	Low	Low	Moderate
Naef 2022 [48]	Low	Low	Low	Low	Low	Low	Low	Low
Gerber 2017 [36]	Moderate	Moderate	Low	Low	Low	Low	Low	Moderate
Gerber 2019 [51]	Low	Low	Low	Low	Low	Low	Low	Low
Suvajdzic 2018 [49]	Serious	Moderate	Low	Low	Low	Serious	Moderate	Serious
Suvajdzic 2019 [50]	Moderate	Low	Low	Low	Low	Moderate	Low	Moderate
Turon 2017 [42]	Low	Low	Low	Low	Low	Low	Low	Low

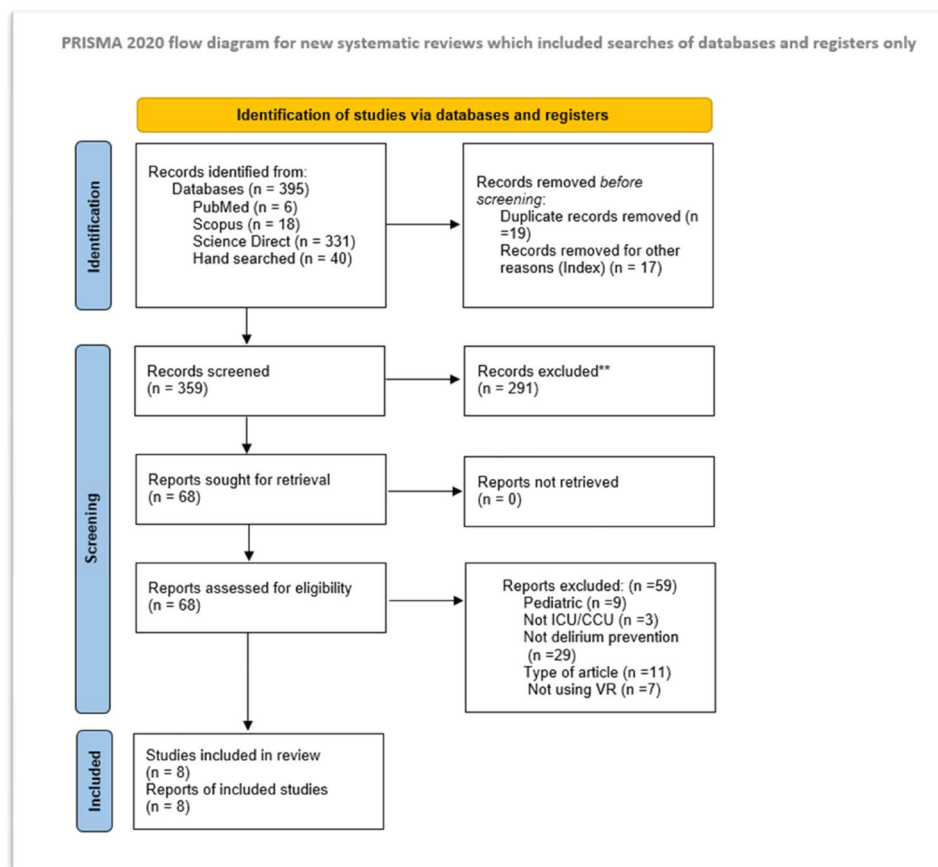


Fig. 1 PRISMA flowchart

20 min VR neurocognitive stimulation sessions, in stimulating attention and working memory, was done by analysis of heart rate of 20 ICU patients, which is considered as an autonomic activity sensitive to cognitive demands. The results emphasize the role of early neurocognitive stimulation in decreasing the incidence and duration of delirium which can improve outcomes in critically ill patients and is considered a feasible, safe and reliable method [42].

Jawed suggests that VR usage may be better for delirium prevention than for delirium treatment and it is a promising technology to help provide patients relief from the noise and stimuli of the typical ICU environment [47]. Also, based on results of using three different stimulation methods for 45 ICU patients in Gerber's study, as an early intervention method directly implemented in the ICU to reduce sensory overload and deprivation, VR nature stimulation may have potential as an early-intervention method directly used in the ICU to reduce sensory overload and deprivation and thus reduce the chance of delirium occurrence [51].

DREAMS project, which consists of VR apps (not specifically made for delirium) with various amounts of Interactivity with the goal of preventing delirium in ICU

patients, found support for the preliminary feasibility of the DREAMS Health VR to prevent delirium in the ICU [50]. Assessment of 10 ICU patients in this study conducted by Suvajdzic resulted in patients remaining CAM-ICU-negative throughout their stay in the ICU (zero rate of delirium) [50].

Application of 5G + VR visiting in ICU during COVID-19 shows that VR system in the ICU can reduce the occurrence of Delirium [52].

Patient outcomes

This section reports any secondary outcomes related to patient well-being, as well as the impact of VR interventions on the patient and provider experience in the ICU.

Secondary outcomes

Vital signs show relaxing effects of VR exposure [36] and target stimulus presentation affected visual exploration behavior (visual search vs. visual processing). Also, investigation of three different virtual reality environments (dynamic, virtual, nature and urban environments) by Gerber, shows nausea, oculomotor problems, and disorientation incidents being close to the minimum of the score scale among 45 ICU patients [51]. Different VR

Table 2 Characteristics of included studies

author	year of publication	country	population	aim of the study	study type	Intervention type	Comparator (if applicable)	outcomes (including Acceptability)	Quality Score based on RoB 2/ROBINS-I
Gerber et al. [36]	2017	Switzerland	37 healthy participants in the ICU (23 female/14 male/20–85 years old)	To test the feasibility and effects of controlled visual and acoustic stimulation in a virtual reality (VR) setup in the ICU	Proof-of-concept study (Single arm prospective repeated measures design study)	- VR for cognitive impairment - Three 2D nature videos, every five minutes in length, played as side by-side videos	N/A	- Vital signs show the relaxing effect of exposure to virtual nature environments - There was no indication of fatigue in visual exploration during VR stimulation and target stimulus presentation affected visual exploration behavior (visual search vs. visual processing) - The VR scenario was well tolerated by the participants and allowed them to fully immerse themselves within the video content - A commercially available VR setup can be adapted to comply with hospital hygiene and safety requirements for use in critically ill patients	Moderate risk of bias
Turon et al. [42]	2017	Spain	20 critically ill adult patients undergoing or having undergone mechanical ventilation for ≥ 24 h	to describe an early neurocognitive stimulation intervention based on virtual reality for patients who are critically ill and to present the results of a proof of-concept study testing the feasibility, safety, and suitability of this intervention.	Proof-of-concept study	- Patients received daily 20-min neurocognitive stimulation sessions when awake and alert during their ICU stay - Physiological data were recorded before, during, and after each session - Safety was assessed through heart rate, peripheral oxygen saturation, and respiratory rate - Heart rate variability analysis, an indirect measure of autonomic activity sensitive to cognitive demands, was used to assess the efficacy of the exercises in stimulating attention and working memory	N/A	- Patients successfully completed the sessions on most days - No sessions were stopped early for safety concerns, and no adverse events occurred - Heart rate variability analysis showed that the exercises stimulated attention and working memory - Critically ill patients considered the sessions enjoyable and relaxing without being overly fatiguing - Virtual-reality based neurocognitive interventions are feasible, safe, well received and tolerable for critically ill patients. - Early neurocognitive stimulation promises to improve outcomes in critically ill patients, potentially decreasing the incidence and duration of delirium, shortening ICU and hospital stays, and ultimately reducing costs while improving cognitive function and quality of life after discharge	Low risk of bias

Table 2 (continued)

author	year of publication	country	population	aim of the study	study type	Intervention type	Comparator (if applicable)	outcomes (including Acceptability)	Quality Score based on RoB 2/ROBINS-I
Suvajdzic et al. [49]	2018	USA	ICU patients	To evaluate VR technology use in delirium therapy in ICU	Qualitative study	Assessing the feasibility of a novel, immersive, intelligent system to prevent delirium	N/A	-discussing the quality of visuals in a VR unit -discussing the Critical components of therapeutic virtual environments	Serious risk of bias
Gerber et al. [51]	2019	Switzerland	45 ICU patients (22 male and 23 female/22–87 years old)	Investigating three different virtual reality environments and their potentially restorative and relaxing effects for reducing sensory overload and deprivation in the ICU	Observational Study	-approximately 80 min for each participant -three different stimulation methods each 10 min in length (dynamic, virtual, nature and urban environments presented inside a head-mounted display and a neutral video on an ICU TV screen) -data was collected by validated questionnaires (i.e., restoration and sickness) and sensors to record physiological parameters	N/A	-In all three environments, nausea, oculomotor problems, and disorientation were close to the minimum of the score scale -higher restorative and relaxing effect in the virtual nature environment compared to the virtual urban environment and the ICU TV screen -three different methods evoked none to minor negative reactions -the highest perceived restorative effect was found in virtual nature stimulation, followed by the virtual urban stimulation and the movie presented on the ICU TV screen -nature VR stimulation had the highest and only significant reduction in respiration rate -nature VR stimulation showed a relaxing effect on heart rate -urban scene sounds may cause sensory overload -previous experiences of patients regarding different environments should be considered -VR nature stimulation may have potential as an early-intervention method directly used in the ICU to reduce sensory overload and deprivation	Low risk of bias

Table 2 (continued)

author	year of publication	country	population	aim of the study	study type	Intervention type	Comparator (if applicable)	outcomes (including Acceptability)	Quality Score based on RoB 2/ROBINS-I
Suvajdzic et al. [50]	2019	USA	10 ICU patients (not intubated) in general ICU of a university-affiliated hospital (20–76 years old)	To evaluate the feasibility of VR-based meditation for preventing delirium in ICU patients	Single arm prospective pre-and posttest design study	- DREAMS project - VR apps with various amounts of Interactivity - Google Daydream with a smart phone - Hand-held Controller with audio via headphone - 10–20 min of cumulative time in VR over two days - VR apps were not specifically made for delirium	standard hospital care	- found support for the preliminary feasibility of the DREAMS Health VR to prevent delirium in the ICU - patients remained CAM-ICU-negative throughout their stay in the ICU (zero rate of delirium) - patients experienced benefits to their pain management and sleep quality, - patients enjoyed and promoted the DREAMS VR platform	Moderate risk of bias
Jawed et al. [47]	2021	USA	15 ICU patients and 21 providers	To ascertain patients' and providers' perceptions of acceptability and safety of VR intervention in ICU	Single arm prospective study- non randomized	-Administering 15-minute session showing a relaxing beach scene with VR headsets and nature sound effects - answers were documented on the paper survey -RASS and CAM-ICU assessments were completed before and after the intervention	N/A	-The majority of patients rated the headsets as moderately to very comfortable -providers in the ICU found the equipment to be well-functioning and easy to use -VR is a promising technology to help provide patients relief from the noise and stimuli of the typical ICU environment -suggests that VR usage may be better for delirium prevention than for delirium treatment -VR intervention is safe, compatible with the ICU infrastructure, and does not interfere with life support equipment -risk of cybersickness in critically ill patients with VR therapy is minimal	Moderate risk of bias
He [52]	2022	China	71 ICU patients	establishing communication between patients and their families	cohort study	-application of 5G+VR visiting in ICU during COVID-19		-The application of the 5G+VR system in the ICU can meet the visiting needs of family members and patients - The application of the 5G+VR system in the ICU can reduce the occurrence of Delirium -VR may reduce the nursing workload and the need for analgesic and sedative drugs	Moderate risk of bias

Table 2 (continued)

author	year of publication	country	population	aim of the study	study type	intervention type	Comparator (if applicable)	outcomes (including Acceptability)	Quality Score based on RoB 2/ROBINS-I
Naef et al. [48]	2022	Switzerland	31 participants (19 former critically ill patients/12 nursing experts)	To determine, based on the opinions of both experts and patients, the optimal type, content, and duration of stimuli for ICU patients during their hospitalization in ICU	Mixed method study	Structured interview	N/A	-providing visual and auditory stimuli to critically ill patients are important for their recovery -nature scenes, including flowers and animals, would be enjoyable -auditory stimulation could last up to one hour, but visual stimulation should not last longer than 10–15 min	Low risk of bias

environments evoked none to minor negative reactions [51]. Nature VR stimulation had the highest and only significant reduction in respiration rate [51]. Nature VR stimulation showed a relaxing effect on heart rate while urban scene sounds may cause sensory overload [51].

DREAMS project shows benefits for patients regarding pain management and sleep quality [50]. Also, risk of cybersickness in critically ill patients with VR therapy is minimal [47].

In Turon's study, 20 critically ill adult patients undergoing or having undergone mechanical ventilation, received daily 20-min neurocognitive stimulation sessions during their ICU stay and successfully completed the sessions on most days [42]. Heart rate variability analysis showed that VR can stimulate attention and working memory which can decrease the amount of ICU and hospital stays, and ultimately reduce costs while improving cognitive function and quality of life after discharge [42].

Patient and provider experience

Jawed's study ascertaining patients' and providers' perceptions of acceptability and safety of VR intervention in ICU shows that the majority of patients rated the headsets as moderately to very comfortable [47]. As for providers in the ICU, they also found the equipment to be well-functioning and easy to use [47]. In Suvajdzic's study, results show that ICU Patients enjoyed and promoted the DREAMS VR platform [50]. Patients considered the sessions enjoyable and relaxing without being overly fatiguing. Therefore, Turon reports virtual-reality based neurocognitive interventions as feasible, safe, well received and tolerable for critically ill patients [42].

Family support is important for ICU patients [53]. He et al. showed in a cohort study with 141 ICU patients, that the application of the fifth generation plus virtual reality (5G + VR) system in the ICU with the purpose of establishing visitation channels for patients and their families during the COVID-19 pandemic, can meet the visiting needs of family members and patients and decrease anxiety and proportion of delirium [52]. VR can also reduce the nursing workload and the need for analgesic and sedative drugs [52].

Adverse events

This section describes any adverse events or unintended consequences associated with the use of VR interventions for delirium prevention in the ICU.

Gerber found no indication of fatigue in visual exploration during VR stimulation [36]. Also Turon's report doesn't show any early stop of sessions due to safety concerns, or any occurrence of adverse events [42]. Across some studies, some minor adverse effects, like isolated cases of transient cyber sickness including mild dizziness or oculomotor discomfort, were noted

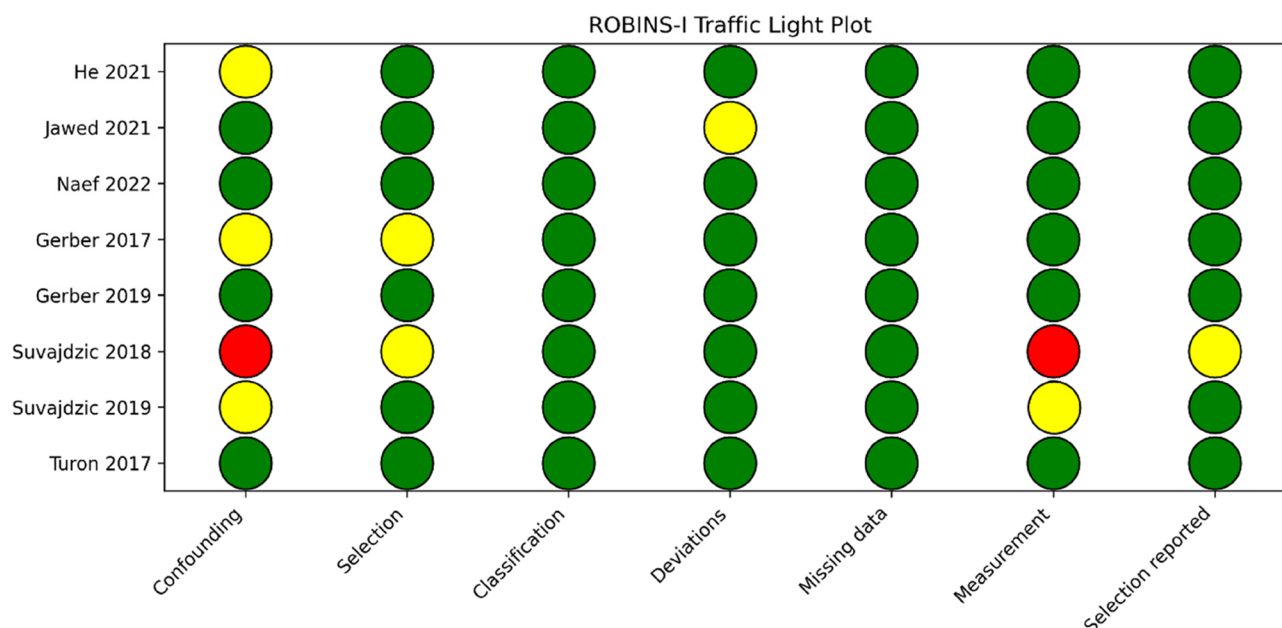


Fig. 2 ROBINS-I Traffic-Light Plot. Green (low risk of bias), yellow (moderate risk of bias), and red (serious risk of bias)

but typically resolved without intervention [48, 51]. No serious adverse events directly attributable to VR were documented. In Turon's cohort, VR sessions were administered to patients on mechanical ventilation without hemodynamic instability or respiratory compromise [42]. However, tolerance varied in patients with higher illness severity, underscoring the need to assess stability before initiation.

Subgroup analyses

This section presents the results of subgroup analyses based on safety concerns, different types of VR interventions, delirium assessment tools and subgroup analysis by intervention parameters.

Safety concerns

Healthcare-associated infections (HAIs) are a major public health burden in ICU [54]. Regarding infection control concerns, a commercially available VR setup can be adapted to comply with hospital hygiene and safety requirements for use in critically ill patients [36]. VR intervention is safe, compatible with the ICU infrastructure, and does not interfere with life support equipment [47].

Different types of VR interventions

Suvajdzic discusses the quality and components of therapeutic virtual environments [49]. Virtual nature environment exposure, was well tolerated by ICU patients and allowed them to fully immerse themselves within the video content [36]. Gerber's study showed that the highest perceived restorative effect was found in virtual

nature stimulation, followed by the virtual urban stimulation and the movie presented on the ICU TV screen. This finding suggests that virtual nature environment provides higher restorative and relaxing effect compared to the virtual urban environment and the ICU TV screen. Urban sounds may cause sensory overload [51]. Naef also agrees that nature scenes, including flowers and animals, would be enjoyable for ICU patients and suggests that auditory stimulation could last up to one hour, but visual stimulation should not last longer than 10–15 min [48].

Previous experiences of patients regarding different environments should be considered for determining the best VR environment for each patient [51].

Delirium assessment tools

Across studies, the definition and measurement of delirium varied, which limits comparability of results. Two studies used the Confusion Assessment Method for the ICU (CAM-ICU), a validated and widely recommended tool for detecting delirium in critically ill patients [50, 52]. In Suvajdzic's DREAMS project, daily CAM-ICU screening confirmed no delirium cases in all participants during ICU stay [50]. He et al. applied CAM-ICU before and after VR visitation, reporting a significant post-intervention reduction in delirium rates [52]. Turon et al. focused on surrogate neurocognitive outcomes (e.g., heart rate variability as a marker of attention and working memory) without specifying standardized delirium criteria [42]. Gerber et al. and Naef et al. did not apply a formal delirium diagnostic tool, relying on clinical observations or indirectly assessing related risk factors instead [36, 48, 51]. Jawed et al. did not specify the tool used,

but prevention effects were inferred from routine ICU assessments [47]. This heterogeneity ranging from standardized screening to informal clinical observation could partially explain differences in reported effects.

Subgroup analysis by intervention parameters

Across the included studies, the VR intervention protocols varied considerably in duration, frequency, and visual content, which may have contributed to differences in observed effects on delirium prevention. Interventions ranged from short sessions of approximately 10 min (Gerber et al.; Naef et al.) to longer exposures of up to 20 min (Suvajdzic et al.; He et al.) [36, 49, 51, 52, 55]. Content also differed as several studies employed nature-themed VR environments (e.g., forests, beaches, mountains), whereas others included urban or mixed scenes. Frequency of sessions varied from once-daily application to multiple or on demand sessions, depending on patient tolerance and ICU protocols.

Studies that used nature themed VR environments such as forest paths, beaches, and mountain landscapes more frequently reported favorable trends in patient mood, relaxation, and in those employing standardized delirium screening, a lower incidence of delirium [36, 48, 50, 52]. By contrast, interventions incorporating predominantly urban or mixed-scene content yielded less consistent effects [42, 51]. However, the small sample sizes and methodological diversity precluded formal statistical meta-analysis to confirm these trends. Interventions exceeding 15 min were not uniformly more effective than shorter sessions of around 10 min, and tolerability appeared to be an important limiting factor particularly in patients with higher illness severity [36, 48].

These findings indicate that visual content of VR stimuli may be a more promising focus for future studies, while duration and frequency may require individual tailoring for feasibility and patient comfort.

Discussion

In this discussion, we evaluate the efficacy of VR interventions in ICU settings, focusing on their impact on patient outcomes, adverse events, and subgroup analyses.

Several studies highlighted in the review demonstrate the positive impact of VR interventions on cognitive function, patient well-being, and overall ICU experience.

Multiple studies indicate that VR interventions may help prevent delirium in ICU patients by targeting various risk factors, particularly sensory imbalances [56]. Naef et al. highlighted the importance of visual and auditory stimuli for recovery, as the absence of delirium is crucial for critically ill patients [48, 56, 57]. Turon's pilot study demonstrated that daily VR neurocognitive stimulation can reduce the incidence and duration of delirium [42], while Navarra-Ventura et al. found that

non-immersive VR can improve cognitive outcomes in critically ill patients [58]. Gerber's research supports the use of VR nature stimulation to mitigate sensory overload and potentially reduce delirium risk [51]. Additionally, Jawed emphasized VR's role in alleviating stress in the ICU environment [47], and Ong et al. noted its effectiveness in reducing anxiety and depression [59].

The DREAMS project and the use of 5G + VR during the COVID-19 pandemic further illustrate VR's potential in preventing delirium, showing a zero rate of delirium among patients using VR [50, 52]. Overall, these findings indicate promising trends that VR could enhance patient care by improving cognitive involvement and providing a supportive environment.

From a neurobiological perspective, immersive and semi-immersive VR may help lower the risk of delirium by working on three things at once: attention control, emotional balance, and the body's automatic (autonomic) functions. The thinking and visual tasks in VR can activate the prefrontal parietal network (brain areas important for planning, decision making, and shifting attention) which can still adapt and change even in later life [60, 61]. In the ICU, these brain circuits often go "quiet," contributing to poor focus and executive problems seen in delirium. VR's calm and structured sensory input may also help steady the autonomic nervous system by lowering overactive "fight or flight" signals (sympathetic arousal) and boosting "rest and digest" activity (parasympathetic/vagal tone), which are changes linked to better thinking and lower delirium rates [62, 63]. In critically ill patients, sustained autonomic dysregulation marked by elevated sympathetic tone and reduced vagal activity is common and has been linked to worse inflammation, cerebral dysfunction, and poor outcomes [64]. By giving both mental stimulation and nervous system calming, VR may address several of the underlying processes that drive delirium in ICU patients.

VR interventions have shown potentially beneficial secondary outcomes related to patient well-being. Gerber reported that VR exposure can positively affect vital signs, including reductions in respiration and heart rates, with minimal adverse reactions [36, 51]. Wong et al. demonstrated the feasibility of VR in reducing anxiety and nausea in pediatric chemotherapy patients [65]. The DREAMS project also indicated benefits in pain management and sleep quality [50], while Goudman et al. noted VR's potential for acute and chronic pain management [66]. Furthermore, Chitra et al. found VR effective in improving sleep quality [67], aligning with findings from studies on VR meditation [68]. Turon's work further supports VR's positive impact on cognitive function, potentially leading to shorter ICU stays and improved quality of life post-discharge [42].

Both patients and providers have expressed favorable views on VR interventions. Patients found the headsets comfortable and enjoyed the sessions, while providers deemed the technology effective and user-friendly [42, 47, 50]. The use of VR for patient-family visitation during the pandemic also reduced anxiety and delirium rates, benefiting nursing workloads [52].

While VR can cause “cyber sickness,” studies did not report significant safety concerns related to VR use in delirium prevention [69]. Controlled aspects of the VR environment can be adjusted to minimize these effects [70, 71]. The implementation of VR in healthcare settings can comply with hygiene and safety regulations, ensuring compatibility with ICU infrastructure [36, 47]. Protocols for cleaning VR devices have been proposed to mitigate infection risks [72, 73].

Effective VR interventions require creating therapeutic virtual environments that promote relaxation and well-being [49]. Krog et al. utilized engaging VR scenes that included nature elements, which may help reduce delirium [74]. Naef et al. emphasized the positive impact of nature scenes on ICU patients, suggesting tailored VR experiences based on individual patient histories to enhance engagement and treatment outcomes [48, 51].

However, translating these promising findings into routine practice presents important challenges. Clinical adoption requires consideration of patient tolerance and safety monitoring. While included studies reported no major safety issues, minor adverse effects such as transient dizziness or headset discomfort occurred occasionally [48, 51]. Cyber sickness is a recognized phenomenon in immersive VR, and while incidence was low in ICU trials, preventive measures such as using nature based, low motion content and adjusting field of view settings can further minimize occurrence [75, 76]. In patients on mechanical ventilation or with unstable vital signs, feasibility depends on careful bedside assessment and the presence of trained staff during sessions [42]. Implementation requires not only access to VR hardware and technical support but also sufficient trained staff to deliver and maintain interventions without disrupting routine ICU workflows.

Additional barriers include strict infection control requirements for reusable headsets and controllers, which some protocols address through disposable sanitary covers and medical grade cleaning procedures [73]. Cost-effectiveness remains uncertain, as upfront equipment costs, software licensing fees, and ongoing maintenance must be weighed against potential reductions in delirium incidence, ICU length of stay, and post-intensive care syndrome-related healthcare use [77]. Feasibility may also vary substantially across ICUs depending on staffing ratios, infection control protocols, and physical space constraints, with resource limited or rural settings

potentially facing greater barriers []. A balanced understanding of these barriers, alongside potential facilitators such as integration into existing delirium prevention bundles, is essential for realistic adoption strategies.

By addressing safety concerns and exploring different types of VR interventions, healthcare providers can better understand how VR technology can be effectively utilized in the ICU setting to enhance patient outcomes and well-being. Using VR as a therapeutic intervention in health care has grown in recent years [79, 80]. While VR appears to be well tolerated in ICU patients and also nursing and medical staff seem to find it feasible, its usage in ICU as a method for prevention of delirium incidence has not been evaluated thoroughly yet.

The findings of this systematic review suggest that VR stimulation shows early promise as a potential intervention for reducing the incidence of delirium among ICU patients. Our analysis of the included studies revealed that VR interventions were associated with improvements in delirium symptoms and improving patient outcomes. As VR could help to reduce the medical and financial costs associated with PICS, VR based technological tools have the potential to become cost effective solutions for hospitals and, in particular, intensive care units. The present findings encourage the introduction of digital therapies as adjuvant, feasible, and safe interventional tools in the ICU, but their integration should be measured, context-specific, and supported by both clinical and economic evaluations.

The heterogeneity in VR protocols (spanning session length, frequency, and content type) highlight the current uncertainty in defining an “optimal” VR intervention for delirium prevention. Our subgroup exploration suggests that nature based content may elicit more consistent positive responses, but these observations come from heterogeneous, underpowered studies. Future studies can test different types of VR content, adjust how long and how often sessions are given, and take patient needs into account to create practical, evidence based protocols for ICU care.

Strengths and limitations

While the evidence supporting the use of VR in delirium prevention is encouraging, it is essential to critically evaluate the strengths and limitations of the studies included in this review to better understand the implications of these findings for clinical practice. This review adds to the growing body of evidence for using immersive virtual reality distraction therapy for patients in the ICU and contributes in providing insights into the potential benefits of integrating VR technology into critical care settings to enhance patient care and improve outcomes.

First, Given the novelty of using VR for delirium prevention in critically ill patients, the review included

only eight eligible studies, none of which were randomized controlled trials (RCTs). All were non-randomized designs which inherently limits the ability to infer causality. In accordance with ROBINS-I guidance, studies rated as having serious or critical risk of bias in one or more domains may substantially distort findings. Such results should be interpreted with caution, and any synthesis considered tentative [46]. Several included studies enrolled very small patient samples (some with as few as 10 participants) further increasing imprecision and limiting generalizability. Considerable heterogeneity was identified across multiple aspects:

- Intervention design: VR modalities, content type, session duration, and frequency differed markedly.
- Comparators: Ranged from standard care to alternative sensory interventions.
- Outcome measures: Varied delirium assessment tools, time points, and definitions reduced comparability.
- Study design quality: Risk-of-bias profiles varied, affecting reliability of pooled inferences.

Future research priorities emerge directly from these limitations:

- Standardize VR intervention parameters (e.g., agree on optimal session duration and frequency) to enable cross study comparison.
- Use consistent delirium assessment tools: adopt validated instruments such as CAM-ICU or ICDSC with harmonized assessment intervals.
- Harmonize patient inclusion criteria: define baseline stability requirements (e.g., RASS – 1 to + 1, hemodynamic stability) to ensure safety and reproducibility.
- Progress to high quality RCTs to explore subgroups (e.g., mechanically ventilated patients versus spontaneously breathing).
- Report implementation details: include headset hygiene protocols, staff training, and adverse event monitoring to support translation into practice.

Due to this clinical and methodological heterogeneity, along with the limited number of high-quality studies, a quantitative meta-analysis was deemed inappropriate. Instead, we conducted a qualitative synthesis to summarize the evidence, focusing on study characteristics, findings, and patient acceptability of VR interventions. This approach aligns with best practices when dealing with emerging topics with limited and diverse evidence, allowing for a comprehensive narrative overview while avoiding overstating the certainty of the findings. Researchers

can use the results of this review for decision-making about recruitment methods for future RCTs on this topic.

Conclusion

Evidence on VR as a preventive measure for ICU delirium remains limited by small sample sizes, methodological variability, and heterogeneity in intervention design, restricting the generalizability of findings. While VR is generally well tolerated by ICU patients and considered feasible by healthcare providers, its effectiveness, as a preventive measure for delirium has not yet been established with high quality evidence. The results of this systematic review indicate that VR stimulation shows promising trends as an intervention to reduce the incidence of delirium in ICU patients, effectively alleviating symptoms and enhancing overall patient outcomes in selected contexts. However, the available studies are small in sample size and vary substantially in methodology, which limits the generalizability of the findings. Future large-scale, multicenter randomized controlled trials are needed to confirm these potential benefits and better define optimal patient selection, intervention parameters, and long-term impacts. Furthermore, VR has the potential to reduce the medical and financial burdens associated with Post-Intensive Care Syndrome (PICS), but such cost effectiveness claims also require validation in robust economic evaluations.

Supplementary Information

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Supplementary Material 1.

Acknowledgements

Declaration: Using artificial intelligence chatbots: We utilized AI-powered tools to check grammar and enhance the academic quality of the text, which was primarily written by the authors.

Authors' contributions

MS collected and analyzed data. ZSM helped with data collection and analysis. AH helped with deciding with disagreements for data collection and helped with analysis.

Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Competing interests

The authors declare no competing interests.

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